Coding & Solution

Date	4 July 2026
Team ID	SWTID-2026-71 47
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

- Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- The solution should address the core problem statement and implement the key functional requirements
- Include all source files, configuration files, and setup instructions needed to run the application.
- Attach or linkto your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/pavanirayapati/AI-ML-and-GEN-AI-Track-Proje ct-Template
Programming Language(s)	Python
Framework(s) Used	Flask
Key Features Implemented	Loan approval prediction, Applicant data analysis, ML model comparison (Decision Tree, Random Forest, KNN , XG Boost), Real-time prediction using Flask
Pending /Incomplete Features	Cloud deployment and additional security enhancements
Setup / Run Instructions	Install Python dependencies, run the Flask application, open the web application in a browser, enter applicant details, and view the loan prediction.

Code Quality Checklist

S.NO	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The Smart Lender application uses machine learning models (Decision Tree, Random Forest, KNN, and XGBoost) to predict loan approval. The best-performing XGBoost model is integrated with a Flask web application for real-time predictions, providing an efficient and accurate solution for credit evaluation.